

Exploring Design Tensions in Countering Microaggressions in Online Communities for Stigmatized Health Support

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Abstract

Many women from diverse cultures seek support in online communities for stigmatized sexual and reproductive health, but often encounter subtle microaggressions that invalidate their experiences and discourage them from seeking care. This study focuses on unmarried Korean women, who face similar challenges in these spaces, often resulting in severe health complications. Through an asynchronous remote focus group with 26 unmarried Korean women, we explored design tensions in developing features to counteract microaggressions in online health communities supporting stigmatized health care. Our findings revealed four key design tensions: balancing content filtering with open communication, promoting personal growth while preserving free expression, providing adaptive support, and encouraging reflection without restricting dialogue. We propose design recommendations to address the design tensions. These tensions, and their proposed resolutions, offer insights for designing provocative prototypes that provide further guidance in creating safer online spaces for supporting stigmatized health care for women.

CCS Concepts

- **Human-centered computing** → **Empirical studies in HCI**;
Web-based interaction; **Computer supported cooperative work**;
- **Information systems** → **Social networks**.

Keywords

microaggression, online communities, sexual and reproductive health, women's health

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1 Introduction

Many individuals turn to online communities for health information and support, expecting them to be safe spaces where they can be free of judgments, relate to others in similar situations, and receive the support they were seeking [25, 33, 37]. Such spaces are especially important in cases where necessary but stigmatized health care could be discouraged [8, 19, 20]. In particular, sexual and reproductive health (SRH) care is a taboo topic in many sociocultural contexts [9, 12, 56, 98, 99]. For example, in Japanese culture, discussions about reproductive organs are considered to violate the code of civilized morality – “sex-related behaviors outside of marriage are thought to be indecent and require silence or euphemism” [32]. In South Africa, cultural and gender norms discourage mothers from discussing sexuality issues with even their own daughters [2]. For Muslim women, marital status dictates the permissible reproductive health care because sexual activity outside of marriage is largely considered taboo [26].

The specific setting we examine in this study is South Korea (hereafter referred to as Korea). Although the Korean government promotes necessary SRH care from an early age, such as supporting 12-year-old girls in receiving the HPV vaccine and counseling, and there have been shifts in sociocultural beliefs to decouple female sexuality from marriage, the stigma prevents many young women from seeking this care, even when they have symptoms [4, 18, 22, 31]. This has resulted in significantly higher cervical cancer rates among Korean women compared to other populations and put them at high risk for severe health complications including pelvic inflammatory disease, ectopic pregnancies, and infertility [1, 18, 31]. These women also lack in-person or offline support from people whom they trust to support and help make decisions with them, such as their mothers or their network of friends and romantic partners [29]. Consequently, they seek refuge in online communities, specifically in women-only forums in NatePann¹, women-only communities in period tracking applications, and general health question and answer forums in Naver Jisikin² [29].

However, in these online spaces, they often face microaggressions—subtle, often unintentional, everyday communications that put down, insult, or invalidate them [24, 29]. Many of the microaggression perpetrators who make these online spaces unsafe are not only anonymous ill-intentioned users, but also fellow Korean women who enforce social norms and often inadvertently commit microaggressions, rather than provide genuine support [6, 29]. The expectation of safety and mutual understanding in these supposedly supportive safe spaces make the experience of microaggressions

¹<https://pann.nate.com/talk/c20024>

²<https://kin.naver.com>

there particularly problematic [103]. In addition, Ryu & Pratt [29] found that these microaggressions do not usually incorporate explicitly insulting words, making them more difficult to detect with current informatics approaches.

In many of these communities, the only two active measures that users could employ to counteract microaggressions are (1) clicking dislike for microaggression comments and (2) flagging them for the moderators to review and delete. Even when allies call out microaggressions, they are often met with the response that they are also too sensitive and the allies become microaggression targets as well [29]. Such lack of options for reducing microaggressions and validating posts and comments amplifies the impact of harmful microaggressions that often go undetected by both automatic and human moderators. [16, 28, 38].

In this study, we examine the design tensions that emerged during an asynchronous remote focus group [7, 12, 21, 27] with unmarried Korean women, who often face microaggressions when seeking stigmatized SRH care. We propose design strategies to address four key tensions in developing features to counteract microaggressions in online communities supporting stigmatized health care. These tensions, and their resolution, offer insights for creating provocative prototypes that can drive large-scale interventions using a design as provocation approach.

2 Methods

2.1 Participants

We recruited 26 unmarried Korean women participants who ranged in age from 20 to 34 ($M = 26.8$, $SD = 3.1$). All participants received primary and secondary education in Korea and were all currently living in Korea. No participants had been married before, and all were currently unmarried. All participants had used online communities prior to the study participation.

2.2 Procedure

This study is part of a larger study with the initiative to explore how to create safer and more supportive online spaces for unmarried South Korean women dealing with the stigmatization of SRH issues. In the larger study, we created an online community where unmarried Korean women participants could reflect on, discuss, and share their SRH experiences and inquiries anonymously through guided and unguided activities for 9 weeks. This study follows the approach of HCI researchers who have utilized asynchronous remote communities as online focus groups [7, 12, 21, 27], leveraging activities within these communities to gain insights into specific design challenges. This study focuses on the two activities specifically designed to gather insights into participants' understanding and responses to microaggressions. These activities were supplemented by interviews that further explored participants' perceptions of microaggression counteraction strategies.

2.2.1 Understanding Microaggression. Since microaggressions and their connection to stigmatized SRH care are not widely understood, with only one study exploring their impact on SRH care for unmarried Korean women [29], we began by educating participants about microaggressions in this context. This activity provided a foundation for examining participants' perceptions of the issue,

learning precise concepts, and encouraging reflection on personal experiences.

A key aspect of our study was encouraging participants to reflect on their experiences with microaggressions, not only as targets but also as potential perpetrators. This approach was informed by the understanding that microaggressions are often committed unintentionally and can occur even among women discussing stigmatized topics such as SRH [29]. Our approach was also informed by the findings of MacLeod et al. [19] and Cooper [5], which highlight the value of using non-victim personas to help participants better recognize problems and identify solutions, rather than limiting their perspective to that of a victim.

This activity began with an introduction to the concept of microaggressions. Participants were first asked *"Have you heard the term 'microaggression' before participating in this study? (yes/no)"* to examine their familiarity with the concept of microaggression and to provide an accurate concept to serve as a basis for the study. Then, microaggression was defined as *"everyday verbal, behavioral, or environmental insults or slights that, whether intentional or unintentional, convey hostile, derogatory, or negative messages toward individuals or groups based on race, gender, sexual orientation, religion, and other factors."*

Using a persona-based approach, relevant personas were provided to illustrate how microaggressions manifest in everyday situations. To deepen participants' understanding, two personas of microaggressions occurring in SRH support provision settings were presented.

- (1) Persona 1 (Target K - Perpetrator J): J, who tells K, who is concerned about a vaginal infection, "That must be tough, but isn't that something that can be okay if you washed up?" Although J provided a solution to the problem, J unintentionally implied that K's issue was due to poor hygiene, thus committing a microaggression.
- (2) Persona 2 (Target K - Perpetrator H): H, who tells K, who is concerned about a vaginal infection (vaginitis), "Don't people get vaginitis often these days? You should've taken better care of it." Rather than offering support or advice, H conveys a judgmental and derogatory message, unintentionally committing a microaggression.

These personas illustrate how individuals, despite having good intentions, can unintentionally convey offensive or judgmental messages. After presenting the personas, we asked participants the following questions for discussion:

- (1) J and H are both microaggression perpetrators. Do you believe both J and H deserve the same title of 'microaggression perpetrators', or do you think this is too harsh?
- (2) Have you ever been a microaggression perpetrator like J and H, even if it was not intentional?

The personas helped participants reflect on how they may have not only been targets or witnesses but also perpetrators of microaggressions. following the guidance of MacLeod et al. [19] and Cooper [5] on using non-victim personas to help participants recognize problems and identify solutions.

2.2.2 Evaluating Designs for Microaggression Counteraction. After the first activity, participants engaged in the second activity which

Table 1: Microaggression Counteraction Design Features

Feature ID	Feature Name	Description
A	Selection of Desired Support	When a user creates a post, they can choose the type of help or support they are seeking. Once the post is published, the selected type of help will be displayed below the post's title. This allows other users to better understand what kind of feedback or assistance the poster is looking for.
B	Microaggression Content Warning	While writing a post, users can select the option to add a "Microaggression Content Warning". If this option is chosen, any comments that contain microaggressions will be automatically blocked. To view the blocked comment, users must click on it and confirm their choice by selecting "Yes" in the warning pop-up. This feature allows users to avoid exposure to harmful content unless they explicitly choose to view it.
C	Display Likes as Hearts Without Dislike Option	This feature replaces the "dislike" option with a heart icon, making only the number of likes visible. Users can click the heart icon to like a post, and the total number of hearts will be displayed beneath the post. This aims to promote a more positive and supportive environment by removing negative feedback options.
D	Automatic Tone Modification for Comments	If microaggression is detected while a user is writing a comment, the system automatically provides a revised version of the comment in a neutral and non-discriminatory tone. Along with the modified comment, an explanation is provided about the changes made. The user is then asked if they would like to accept the revised comment. If they select "Yes", the revised comment will be posted. If they choose "No", the original comment will be posted as written.

focused on evaluating various designs of online community features that aim to counteract microaggressions. These features represented four potential solutions, each addressing different aspects of reducing or mitigating microaggressions and creating more supportive communities (Table 1). To minimize the cognitive and emotional burden on participants, we opted for a structured evaluation. Previous co-design studies on sexual and reproductive health with unmarried Korean women revealed that introducing the concept of microaggressions, asking participants to reflect on their roles as perpetrators, targets, or allies, and generating counteraction ideas on the spot was challenging [30]. By providing four potential solutions, we ensured that participants could focus on thoughtful critique and refinement, rather than the more cognitively and emotionally demanding task of generating ideas [30].

Participants were provided with a visual diagram illustrating these solutions (Figure 1), labeled A through D, and asked to review and answer questions related to each feature. They were asked to critically evaluate the effectiveness of each approach and how it fits into the larger goal of creating a more inclusive and respectful environment.

Following the review of the design features, participants were asked to assess their perceptions of the effectiveness and potential usage of each feature. The Likert-scale questions were scored on a range from 0 to 4, where 0 indicated "not at all" and 4 indicated "very likely." The questionnaire consisted of four Likert-scale questions asking the likeliness of using each feature and one Likert-scale question asking the need for explanations for feature D. These questions allowed participants to express the likelihood of their engagement with each feature, as well as the perceived necessity of specific components. The questionnaire had one final open-ended question: *Among the features (A), (B), (C), and (D), which do you believe would be most effective in addressing unintentional but hostile, contemptuous, or negative messages? Which feature(s) do you think would not be helpful? Could you explain your reasoning? Are there*

any suggestions on how the less helpful features could be improved? There is really no right or wrong answer.

2.3 Analysis

We audio-recorded the evaluation interviews and used a professional transcription service to have the recordings transcribed verbatim. The first and second authors co-coded three transcripts and the responses for the two activities with the ATLAS.ti software using an inductive approach. Second, to derive codes representing dominant concepts in the data, the second author clustered related codes into overarching categories using an affinity diagramming approach, wherein she arranged coded excerpts into clusters according to similarity. To validate the coding scheme, she iterated on affinity diagramming with the first author. Third, with the finalized codebook, both authors coded the remaining transcripts.

3 Findings

In this section, we discuss four main tensions that emerged between users over the issue of counteraction to in-group microaggressions. 24 out of 26 participants recognized microaggressions as problematic and 23 participants acknowledged that they had unintentionally perpetrated them. Many participants expressed a desire to share their narratives freely in a safe and non-judgmental online space. However, there was a wide variation in how and to what extent these safe online communities should be created. We observed four main tensions emerging: (1) emotional comfort vs. complete transparency, (2) pre-defining assistance vs. flexibility in needs, (3) education and support vs. restriction and suppression, and (4) direct correction vs indirect feedback.

The screen the user sees while writing a post

Title

Post

Please select the types of support you would like to receive.

- ☒ Information Support
- ☒ Emotional Support
- ☐ Women-Only Assistance
- ☐ Expert Help
- ☒ Discussion (All opinions welcome)

Microaggression Content Warning

☒ Yes
☐ No

Post

(A) Selection of Desired Support
(B) Microaggression Content Warning

The screen the user sees after publishing a post

Concerns about HPV Vaccines and OB-GYN Visits

Writer 2023.11.07

Information Support Emotional Support Discussion Views 1,632

Hello, I'm a 31-year-old woman, and I've recently become interested in the HPV vaccination. I think it would be helpful if my boyfriend also got it, but I guess he doesn't really want to :/ He asks if he really has to go to an OB-GYN and wonders if I don't trust him... I'm a little scared to go alone though... I've never been, so if I go, will I have to sit on the gynecological examination chair with leg stirrups? Will I be scolded for not coming earlier? Will I get strange looks? I've been worrying about this for a while, and I'm finally posting about it.

89

Comments

Recent
Most Likes
Most Dislikes

10

Microaggression Content

(D) Automatic Tone Modification for Comments

Is it really something to stress over that much?
You could just go instead of overthinking this trivial stuff.

Reply

[A comment with a neutral and non-discriminatory tone]
[Explanation of the tone modification]
Would you like to change it as described above?

Yes No

(B) Microaggression Content Warning

Microaggression Content Warning

This comment contains microaggressions. Do you still want to view it?

Yes No

Click "Yes" to reveal the content of the comment

Is it really something to stress over that much?
You could just go instead of overthinking this trivial stuff.

Figure 1: Microaggression Counteraction Design Features. This figure shows how each feature is implemented in a hypothetical online community. In the top section, the user can configure Feature A (Selection of Desired Support) and B (Microaggression Content Warning) before posting. The bottom section shows the window that appears after the user publishes the post for Features A and B. Features C (Displaying Likes as Hearts Without Dislike Option) and D (Automatic Tone Modification for Comments)'s usages are also demonstrated in the bottom section of the diagram. The figure has been translated into English but was presented in Korean to the participants during the study.

3.1 Emotional Comfort vs. Complete Transparency

The balance between emotional comfort and complete transparency presents a significant challenge in online interactions. On the one hand, features that block or filter microaggressions are designed to create a safer and more supportive environment, fostering open communication. On the other hand, the pursuit of transparency and unfiltered feedback can conflict with this goal, as some users value the ability to address all comments directly, regardless of their potential discomfort.

Several participants shared personal experiences with microaggressions, explaining how they hesitated to participate in online communities for fear of judgment or ridicule. P8 mentioned being dismissed in an online forum where their SRH concern was "labeled as overreacting", which left them reluctant to post again. These experiences made features like A (Selection of Desired Support) and B (Microaggression Content Warning) particularly appealing, as participants felt these features could provide a sense of control and emotional safety.

A notable number of participants emphasized the significance of features designed to shield them from microaggressions and negative interactions, such as feature A (Selection of Desired Support), feature B (Microaggression Content Warning), and feature C (Display Likes as Hearts Without Dislike Option). Many participants highlighted the importance of features that enabled blocking and selective viewing of messages. They believed that minimizing exposure to microaggressions could significantly mitigate their impact.

Feature B's limiting microaggression comments function would help block unpleasant messages within the community - P7

Some participants also noted that feature B would make it easier to share concerns comfortably. They stated that such a feature could foster a more supportive and less stressful online environment, allowing individuals to express their worries more freely.

If microaggression comments are only visible when the user chooses, it would allow for freer and more comfortable online discussions and advice-seeking without the fear of encountering harmful content. - P20

Conversely, concerns were raised regarding feature B. P14 acknowledged the unpleasant nature of microaggressions but emphasized a desire for raw, unfiltered comments rather than filtered or "sugarcoated" responses. This perspective suggests that the push for emotional comfort may not align with the preferences of all users.

Although I find microaggression expressions unpleasant, they do not need to be filtered to the extent that I wouldn't want to see raw, unfiltered responses. - P14

3.2 Pre-defining Assistance vs. Flexibility in Needs

The dilemma between pre-defining assistance and allowing flexibility in addressing user needs represents another significant challenge. Feature A, which pre-defines the type of assistance users

might require, is highly valued by those with a clear understanding of their needs. However, for those who are uncertain about their needs, this approach can be problematic, potentially leading to misdirected assistance and complicating the help-seeking process.

Many participants shared examples of how microaggressions took an emotional toll, often in the form of unsolicited advice that felt dismissive or invalidating. P10 recalled receiving a comment that suggested their health concern was not serious enough to warrant advice, which led them to disengage from further discussions. Feature A was considered helpful in envisioning a more structured approach to soliciting the support they genuinely sought.

Unsolicited advice dismissed my concern as not serious enough, which was emotionally exhausting and made me disengage. Clear support options like feature A (Selection of Desired Support) could help avoid this. - P10

Participants expressed high expectations for feature A's practical implementation to create a more supportive environment, anticipating that a supportive environment would lead to more desirable and positive responses. In fact, a couple of participants emphasized that understanding the intentions of others is an important factor in being able to engage in, and avoid, microaggressions. They noted that "when you have a clear understanding of the support the other person is looking for, you're more likely to respond appropriately (P20)," and therefore less likely to make unintentionally hurtful comments. Participants agreed that this understanding would help them tailor their responses to the needs of others, creating a more supportive environment. Additionally, knowing in advance what kind of help the poster wants allows users to reconsider the potential negative impact of their comments.

If you know in advance what kind of help the poster wants, it naturally gives you a chance to reconsider whether your comment is problematic. - P5

Accordingly, we were able to see that feature A is expected to have a positive expectation on people who don't want to be negatively affected by microaggressions, but also on people who don't want to inadvertently commit microaggressions.

However, P17 pointed out a potential downside to feature A, which could potentially lead to misdirected assistance. They mentioned that while A might be beneficial for those with some understanding of their needs, it might complicate the process for those entirely unsure.

I think feature A is good, but pre-defining the type of assistance might limit the help received (especially if I don't even know what kind of help I need). While detailed categorization is useful for those with some knowledge, for those who are completely ignorant, it could lead to misdirection and difficulty in choosing, so I rated it lower than other features. - P17

3.3 Direct Correction vs. Indirect Feedback

The tension between direct correction and indirect feedback highlights different approaches to addressing microaggressions within the community. On the one hand, direct correction through features like D (Automatic Tone Modification for Comments) aims

to address microaggressions at the source by providing immediate, explicit feedback and educational approaches for users. This perspective focuses on directly correcting potentially harmful comments and guiding users toward more respectful communication. On the other hand, indirect feedback mechanisms, such as the dislike button (opposite to feature C), offer a subtler form of response by providing feedback on posts, without directly confronting the user. This method allows users to address microaggressions within the content itself, potentially fostering more organic self-correction and awareness.

Participants like P16 shared stories about how a blunt correction they received online felt more like public shaming than constructive feedback, which deterred them from posting further. However, they acknowledged that indirect feedback mechanisms, such as subtle nudges or community guidelines, could provide a more effective way to encourage reflection without alienation.

Blunt corrections felt like public shaming and made me hesitant to post again. A subtler approach would have encouraged reflection. - P16

Many participants expressed a desire for impartial self-assessment to avoid being involved in microaggressions and found features C and D useful for this purpose. By providing explanations on tone adjustment through feature D, users could gain insights into how their words might be perceived by others, encouraging thoughtful self-evaluation and fostering more respectful conversations. Additionally, participants thought that replacing the lie/dislike mechanism with feature C would help raise awareness of the harms of microaggressions within the community. They mentioned that showing the collective sentiment of the community towards a comment could aid users in evaluating their behavior more effectively.

The core of online communities is "convenience" and "anonymity", and using hearts is quite simple. I think feature C (Display Likes as Hearts Without Dislike Option) serves as a mechanism to show how rude and provocative one's or others' words are based on the quantified number of hearts. - P24

A fair number of participants underscored the importance of having a system that provides immediate suggestions on how to convey their messages in a more considerate way. **"It is not enough to just recognize microaggressions but to need a direct guideline on how to actually correct this,"** as P24 noted. Participants pointed out that **there are currently not many spaces or platforms that suggest ways to reduce discriminatory messages**, so they need platforms that offer actionable feedback to help users adjust their language to avoid microaggressions.

I think feature D (Automatic Tone Modification for Comments) is a very fundamental countermeasure in that it immediately provides information about microaggressions to those who use them and provides an educational opportunity to correct their own expressions. - P6

Despite the positive impacts, there were notable concerns regarding features C and D. For feature C, one participant pointed out that removing the dislike option for posts leaves no way to address microaggressions that may arise from the posts themselves.

P20 expressed a desire "to click the dislike button if the content of a post is biased or offensive," highlighting the need for a way to handle such content. P20 raised concerns that, in addition to direct means such as comments to correct or criticize the post, indirect means like the like/dislike buttons are necessary to address microaggression content. Notably, although she was not an active user posting content before the experiment, she frequently browsed the community as an indirect community user, indicating that their means of expression should be respected.

3.4 Education and Support vs Restriction and Suppression

Lastly, balancing the goals of education and support with the need to protect free expression presents a complex challenge within online communities. While participants underscore the importance of educational features, such as features B (Microaggression Content Warning), C (Display Likes as Hearts Without Dislike Option), and D (Automatic Tone Modification for Comments), to prevent unintentional microaggressions and foster a more inclusive environment, concerns arose regarding the potential for these same features to overreach. Restricting the tone of discourse too much could inadvertently suppress diverse viewpoints and minority opinions, which are crucial to the community's vibrancy and openness. This tension highlights the difficulty of fostering a supportive space without compromising the community's foundational principles of open dialogue and diverse perspectives.

Many participants shared their apprehension about becoming desensitized to microaggressions due to repeated exposure. They explained that they had encountered posts with subtly harmful language so often that they initially did not recognize it as problematic. Feature B was considered crucial for breaking this cycle and maintaining a constructive environment.

After seeing so many posts with subtle microaggressions, I started to overlook how harmful they were. It's like I became desensitized without realizing it. A feature like a content warning could stop this cycle and help people recognize and avoid repeating these behaviors. - P19

Several participants claimed that feature D would help users better understand the microaggressions they might cause unintentionally. This design was deemed effective in preventing microaggressions as it provides users with an opportunity to think again before posting a comment. Participants emphasized **the importance of being given the chance to recognize and evaluate their own behavior from an impartial perspective to avoid causing microaggressions**. They also highlighted that feature D would be a fundamental way to reduce microaggressions by providing users the opportunity to be educated on a better way to support fellow unmarried women.

I think feature D (Automatic Tone Modification for Comments) will be very effective in preventing microaggressions as it helps discover the problems in writing from an objective perspective. - P6

Additionally, several participants expressed their hope that the community would become a supportive and positive community

in the long term through features C (Display Likes as Hearts Without Dislike Option) and D (Automatic Tone Modification for Comments), which encourage self-reflection and education. These features are seen as features that allow users to critically assess their own comments and behavior, helping them understand how their words might affect others.

Even though you did not intend to attack someone, harmful emotions can be expressed without noticing it. However, if an explanation can be presented with consent, it will likely help purify the community in the long term. - P17

Additionally, the prevention of microaggression behavior alignment was a prominent theme in the findings. Feature B (Microaggression Content Warning) was considered by participants to help prevent users from unknowingly aligning with negative behavior. **Regardless of understanding what kind of help is wanted and what tone is desired for comments, if a user's comment differs from others, "they could unconsciously align with problematic comments due to being swept up in others' microaggressions and negative comments,"** as P6 expressed concern. And, many participants emphasized that feature B would help prevent this. This approach would also **mitigate the negative loop between exposure to and utilization of microaggressions, reducing the likelihood of users perpetuating such behavior.** Participants regarded repeated exposure to microaggressions would lead to the risk of desensitization or normalization of such behavior. By preemptively blocking these issues, feature B would prevent the spread of such negative behaviors within the community, playing a crucial role in ensuring interactions are positive and constructive.

Feature B (Microaggression Content Warning) helps prevent the vicious cycle of becoming accustomed to microaggressions and using them again. - P19

On the other hand, a few participants expressed concerns that features C and D might overly restrict the online community or suppress minority opinions. The community was initially valued as a space for sharing diverse perspectives and learning from one another. Participants feared that imposing too many restrictions on the tone of comments to prevent microaggressions could undermine the community's core purpose. They advised that while it is beneficial for users to reflect on and learn from microaggressions, this process should not be overly restrictive or stifling. Several participants strongly believed that while features C and D could effectively reduce microaggressions and guide users toward more considerate behavior, the approach should not obscure the essence of what the user wants to convey beneath the tone.

(Feature D) would take away the freedom of speech that I personally think is most important in an online community. Also, I'm not sure how this feature would work, as the tone of voice is very contextual and acceptable to different individuals. With feature D, users would be too stifled, and the online community would not be able to function properly. - P14

4 Discussion and Implications

Several challenges shaped our approach and contributed to identifying the four key design tensions in the development of microaggression counteraction features for online communities supporting stigmatized SRH. First, since microaggressions and their connection to stigmatized SRH care discussions are not widely understood [29], we began by providing an opportunity for participants to learn and reflect on microaggressions through the first activity. This activity provided a foundation for participants to learn precise concepts and reflect on how microaggressions could be unintentionally caused by fellow women, highlighting the complexity of harm within the community.

Second, we encouraged participants to reflect on their roles as unintentional perpetrators of microaggressions, rather than assuming they were "innocent targets". This reflective process aligns with MacLeod et al. [19] and Cooper [5]'s findings that being a victim does not inherently equip one to identify solutions. Recognizing their roles allowed participants to offer deeper insights in their design evaluation.

Finally, we minimized the cognitive and emotional burden on participants by having a structured evaluation of four design features. As participants in other co-design studies on microaggression counteraction for stigmatized health topics faced emotional and cognitive challenges in generating ideas on the spot while understanding and reflecting on new and stigmatized topics [30], this structured approach made participation more comfortable and helped uncover nuanced tensions.

Building on the four key tensions uncovered in the study by addressing the challenges we faced, we propose design directions to resolve these tensions. These tensions, along with their proposed resolutions, offer valuable insights for creating provocative prototypes that can drive large-scale interventions using the design-as-provocation approach.

First, balancing features that allow blocking and selective viewing of emotionally harmful microaggressions with the desire for unfiltered comments is crucial for ensuring a supportive and inclusive online experience. To mitigate this tension, we recommend **providing users with customizable settings that allow them to choose their level of exposure to unfiltered content**, giving them the flexibility to tailor their experience based on personal preferences while maintaining a sense of safety from not being exposed to microaggressions without protection [15].

Second, providing relevant, adaptive support is essential to avoid misdirected assistance. To mitigate this, we suggest **incorporating flexible guidance features that can adapt over time based on user engagement** [11]. For example, users who are uncertain about their needs could start with broad options but later refine their preferences based on feedback from the system or community, ensuring that they receive appropriate support without feeling limited.

Third, it is essential to encourage education and self-reflection without limiting open dialogue or freedom of expression. To address this, we first **propose a hybrid approach that combines direct correction mechanisms, such as tone modification tools, with more subtle forms of indirect feedback, like community ratings or flags of microaggression comments.** We also

recommend using educational prompts that provide feedback without forcing users to modify their behavior, paired with optional reflective tools that invite users to think about the impact of their words. This way, users are encouraged to learn and grow from their mistakes in a non-confrontational way, while ensuring that harmful content is addressed without overly restricting free expression. [9, 13, 14, 17, 23]. There is a lack of clarity in currently available explanations for why a post or comment is removed which leaves users feeling frustrated [13]. Advocates of a more educational approach [9, 14, 17, 23] align with our view that moderation systems can serve as educational tools, fostering growth in perpetrators by encouraging reflection on their behaviors and their impacts.

Our recommendations aim to guide the design of microaggression counteraction tools in online safe spaces. However, this study's focus on unmarried Korean women presents limitations. Future research should extend these insights to other populations facing similar stigmatized health challenges. Another key limitation of our study is the challenge of balancing the designers' values with participants' preferences when making design implications. In navigating the "right" approach to resolving tensions between directness, transparency, and flexibility, we acknowledge our unintentional leaning towards our own perspectives. However, these very tensions and their proposed resolutions offer valuable insights into the design of provocative prototypes (provotypes) [35]. Provotypes serve as experimental tools that challenge users' assumptions and social norms, allowing designers to assess whether their designs, based on user needs and the proposed resolutions for design tensions, were truly desired by and beneficial to the users [3, 10]. By observing how participants engage with these prototypes, we can gather feedback on whether our design approach effectively addresses the identified tensions, providing a clearer understanding of whether the proposed solutions were suitable or if further adjustments are needed [3, 10].

Our study thus serves as a critical first step toward applying the design as provocation approach, with the tensions identified here forming the basis for future provotypes. Other research on marginalized communities supports this direction, showing how provocative designs that prompt reflection and further challenge users' assumptions and social norms lead to more effective solutions than initially preferred options [34–36]. Our study similarly lays the groundwork for using the design as provocation approach to create safer online spaces for stigmatized groups.

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